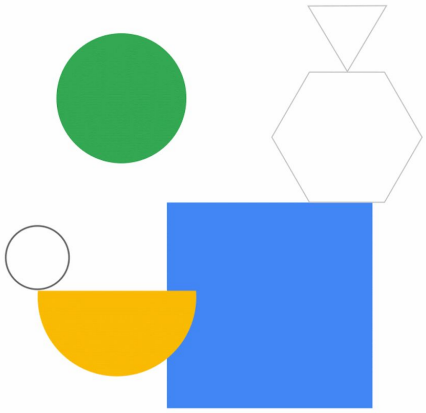
Google Cloud

Migrating to BigQuery



The main objective of this course is to show you, a Google partner, when and how to migrate a data warehouse system to Google Cloud. In this module, we'll talk about migrating the core of the data warehouse system – the data warehouse itself.

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Objectives

01 Assess an existing data warehouse and develop a strategy to migrate it to BigQuery

02 Describe best practices for migrating existing data warehouses to BigQuery

03 Identify key resources, tools, and partner assets for migrating to BigQuery

04 Load and query data using the BigQuery web user interface and the BigQuery CLI

05 Identify resources to translate product-specific SQL queries to BigQuery Standard SQL

Google Cloud

There are a lot of factors to consider in any migration, but a data warehouse is more complicated because in many cases, the data warehouse will have morphed from something else. So understanding the architecture requires a lot of detective work.

- The most important phase of the migration is discovery, so we'll spend quite a bit of time learning how to assess the current system and how to create a roadmap to get to the new, modernized one.
- Google has participated in many migrations, so we can take the lessons learned from those and apply them as best practices in your migrations.
- You aren't alone in the migration process. There are tools such as BigQuery Data Transfer Service to facilitate the migration. There are also other Google partners who provide migration tools or can provide consulting advice during the process.
- In the lab for this module, we'll explore how to load and query data using the BigQuery web user interface and the BigQuery CLI.
- Although SQL is an ANSI standard language, many vendors (such as Teradata) include additional features and functionality to make their version of SQL more useful. The level of effort for converting the SQL depends on how much product-specific SQL was used by the customer, so it's important to scope this during discovery.

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01 Migration Phases

02 Security

03 Google Cloud Data Warehouse Architecture
Lab: Google Cloud Fundamentals: Getting Started with BigQuery

04 Post Migration

05 User Adoption

06 Summary

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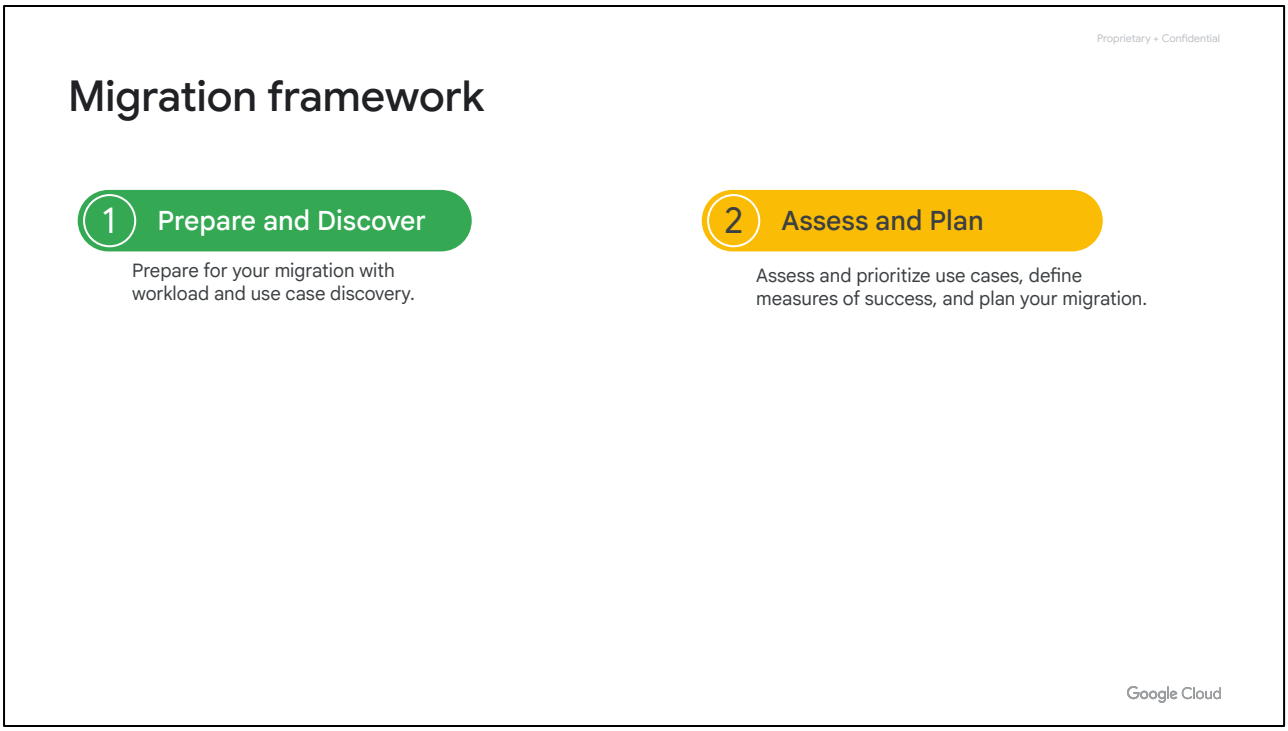
To migrate an existing legacy data warehouse, you should have some idea of the process involved, so let's look at common phases of a data warehouse migration.

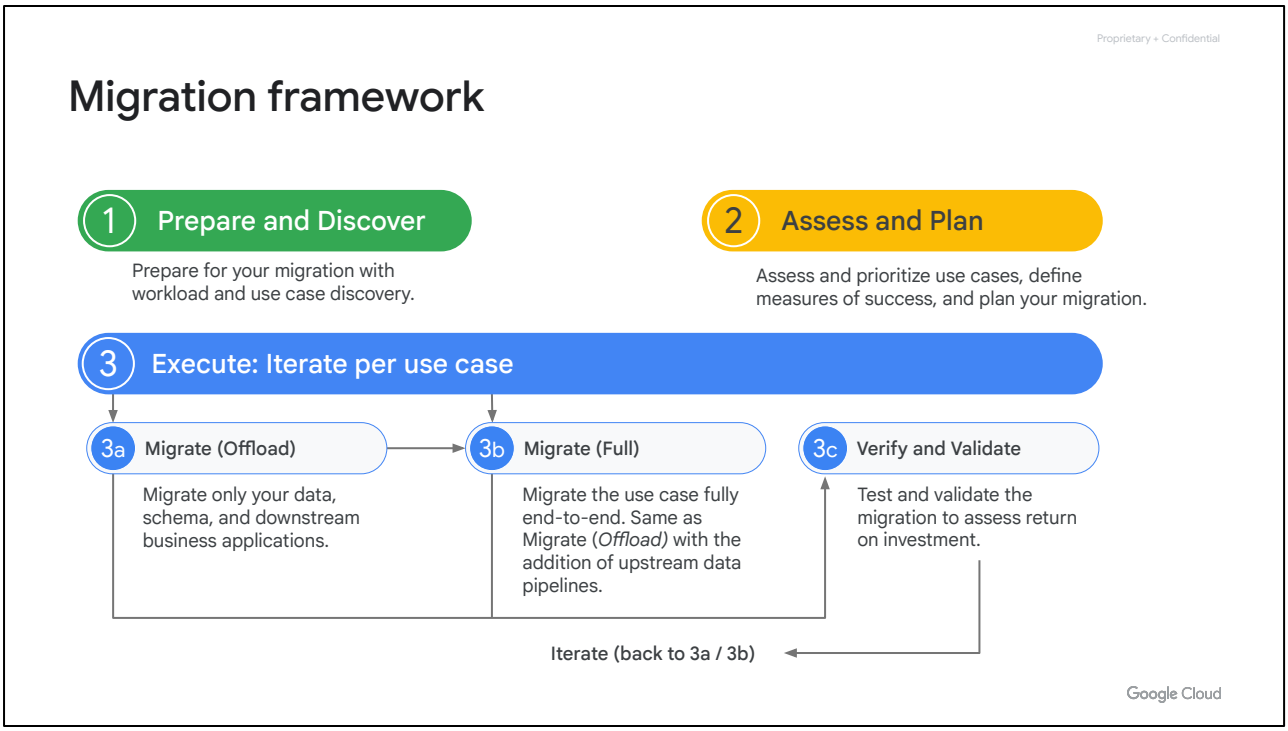


Undertaking a migration can be a complex and lengthy undertaking, so you should use a framework to organize and structure the migration in phases. I'll share one possible framework that you might choose to adopt or that you could use to augment an existing framework in an environment. I'll start by providing a high-level overview of each component in the overall framework and then discuss each of them in more detail once you see how they fit together.

The initial phase focuses on **preparation and discovery** whereby you identify the current configuration and discover the existing use cases. For this discussion, a *use case* consists of all the datasets, data processing, and system and user interactions required in order to achieve business value, such as tracking sales volumes for a product over time. It's important to conduct the analysis with an eye toward the expected benefits of the migration. These can include performance gains and reductions in total cost of ownership (TCO). It will be difficult to justify the migration if it won't provide any measurable benefit. You should involve as many stakeholders as possible during this phase to ensure that you gather a complete set of information and also to start getting buy-in for the system.

The **assess-and-plan** phase takes the output from the prepare-and-discover phase and uses it to plan for the eventual migration. It's necessary to determine what the customer has in place, however, you don't want to simply duplicate that because the current state may include processes that aren't used or that don't work. You should also consider whether or not schema changes are necessary or desirable. While schema changes might improve performance, they will add time and cost to the

project and the customer may not want to make that trade-off. When designing a new system, a proof of concept can be useful both to test the feasibility of the approach and also to demonstrate results to the customer. If you aren't comfortable with some part of the process you might want to include a migration partner to help with that portion. Identifying measurable success metrics is crucial – otherwise how can you be sure you're finished? This will also ensure that you and the customer agree as to what will be delivered and how it will work.



Unless your data warehouse is very small or has limited use, it's a good idea to break the migration into bite-size pieces and **iterate** through them. The pieces can be a single use case or a set of dependent use cases typically called a workload. Each iteration has two main components.

Depending on the size and complexity of the use case, you might decide that, rather than conduct a full migration, the best approach is to leave the current data warehouse in place and use Google Cloud to **offload** part of the workload - perhaps just the end users who consume the data. This approach migrates the schema and data into Google Cloud but leaves the existing ETL pipelines in place. It adds a process to keep the Google Cloud copy of the data synchronized with the existing data warehouse.

Alternatively, your migrated use case may include the **full** end-to-end process. This approach migrates the ETL/ELT processes in addition to the schema and data, so it doesn't require synchronization of the legacy data warehouse with the Google Cloud version. Both approaches have the same foundational requirements, such as setting up networking and security, as well as mechanisms to monitor systems and potentially create alerts under defined conditions. You need not use the same approach for all use cases but generally an offload of a use-case is eventually followed by a full migration.

Each bite of the migration should be **verified and validated** regardless of the migration approach. This phase demonstrates that the migration meets, or exceeds,

the objectives defined in phase 2.

Now that you have the big picture of the overall process, let’s look more closely at how to complete each phase.

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Phase 1: Prepare and discover

Usage patterns

- What kind of usage patterns do you encounter?
- Do you perform high frequency DML statements?
- Do you have sub-second query response time requirements?
- Do you use any specific language or SQL functions?
- Does the environment rely heavily on stored procedures, macros, or triggers?
- Do you have data enrichment needs?
- Do you have data availability SLAs in your environment?

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As part of the discovery process, you need to understand how the current system is being used to ensure that the functionality and performance is matched or exceeded in the migrated data warehouse.

For example, if the current data warehouse uses stored procedures to perform ETL, they must be converted to BigQuery scripts or the ETL process modified to use other services such as Dataflow. The nature and scope of changes required in the migrated data warehouse directly impact the timeline and cost.

This isn’t a definitive list of questions, but represents the type of questions you should ask to help you determine usage patterns.

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Phase 1: Prepare and discover

Data, data sources, and patterns (For each schema)

- How many schemas?
- How many tables?
- What is the size of the largest tables (transactions and volume)?
- What is the primary data model?
- What is your data retention policy?

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Data is the central part of any data warehouse system, so you must understand the scope of migrating it. Scope includes the number of entities, volume of data, and complexity of the data model. If there's a retention policy, you must document the details; different data may be retained for different periods. You might also ask the customer why they don't retain the data. If cost is an issue, you might provide additional value in the migrated data warehouse by retaining this data at a very low cost.

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Phase 1: Prepare and discover

Data processing

- What source systems feed the Instance(s)?
- Describe the data lifecycle objectives.
- What percentage of the data is structured? Semi-structured? Unstructured?
- What ETL tools are used for data loading?
- Do you have a data governance/quality/lineage strategy?

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While data is the central part of the data warehouse, it's not useful if it becomes stale. So it's important to discover all of the sources and processes that feed the data warehouse. If the customer's current ETL tools support BigQuery as a sink, you can leverage existing processes to save money and time. However, they may want to eliminate those tools to get a complete cloud solution, so don't presume they want to keep the existing tools – ask!

Some customers want or need to keep track of the lineage (or history) of data, sometimes referred to as change data capture or CDC. If this is a requirement, it must be documented to ensure the migrated system provides the required information.

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Phase 1: Prepare and discover

Audit and compliance

- How is access restricted and audited? Where and how are each of these managed?
- What is the finest grain of access control that you have implemented (table, view, row/column)? If row/column level, how have you implemented it?
- Can you describe data governance policies that apply to these environments?

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Identifying who has access to what data, and for which operations, is crucial to prevent unauthorized access and also to avoid frustrating end users trying to create or run analyses. Security can be applied at many levels, so you must discover the granularity of access required.

The system might also need to track who has accessed the data: both reading and writing.

Some industries have government-mandated compliance, and most companies have some type of governance policies, both of which must be identified during discovery to avoid penalties.

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Phase 1: Prepare and discover

Testing

- How are test environments managed today?

Testing is often an afterthought, so it's possible that the customer doesn't have a testing environment. If not, this is an opportunity for you to provide value by suggesting the implementation of a testing environment and methodology to employ it. This environment will be used during the migration, so it provides multiple benefits. If the customer does have a testing environment, you should understand how they use it, but you'll probably not duplicate it in Google Cloud.

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Phase 1: Prepare and discover

Users

- Which teams use the current data warehouse?
- What are your most important analytics application areas?
- What tools do you interact with?
- Do you work directly or through BI tools or dashboards on top?

Google Cloud

The reason a data warehouse exists is for the benefit of the users. Don't forget that users aren't a uniform group of people but represent a variety of potential use cases. The data warehouse administrator might be able to identify the users, but you should ask around to make sure you don't miss anyone.

Most dashboard, reporting, and analytical tools support BigQuery as a source, which you can use to minimize disruption to users. However, any changes to the schema may require users to revise their queries. You should also find out if users are unhappy with their current tools or if they find them too costly, so that you can suggest alternatives.

It's important to speak directly to the end users if at all possible. They are the only ones who really know how they are currently using the data and, more importantly, things they might wish they could do…..

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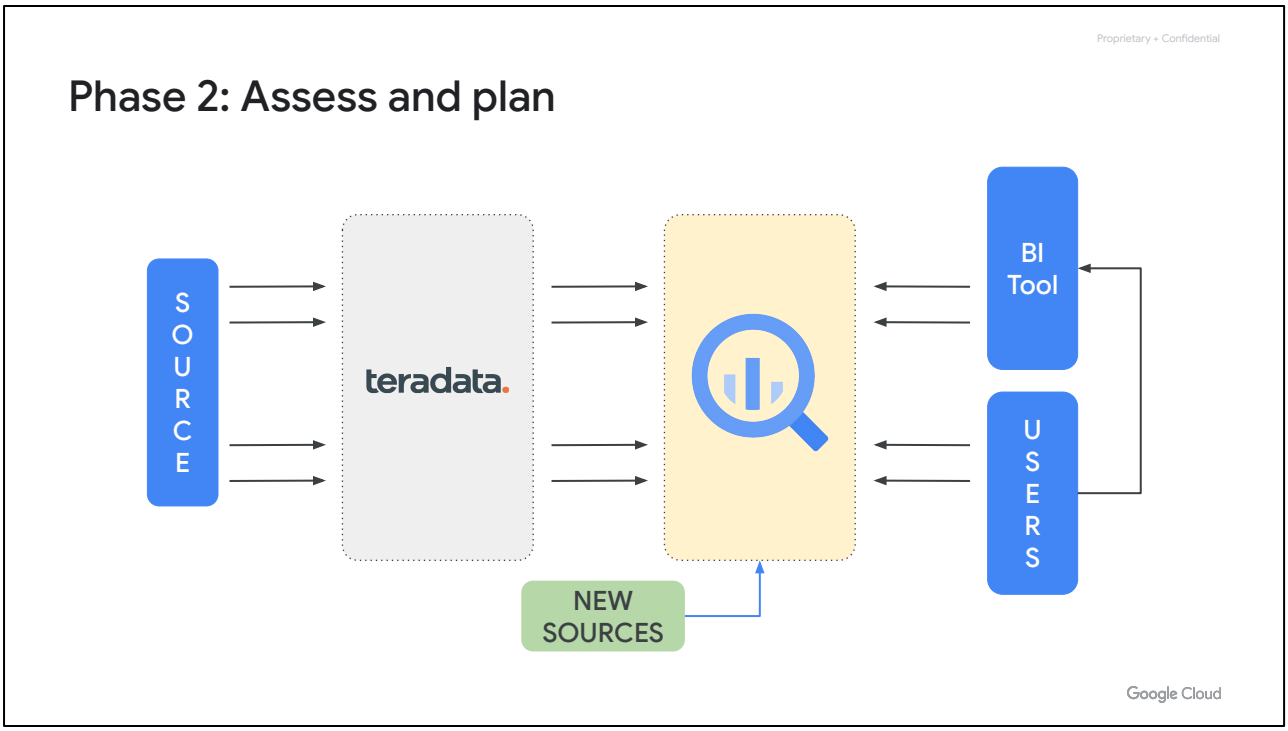
Phase 1: Prepare and discover

Business users

- What data do you load into your warehouse(s) and data lake(s) in real-time?
- What is the extent of the real-time analysis or decision-making use cases you support? How real-time would you like it to be?
- What kind of pain points do you observe?
- Do you have access to all the data you need, and in a timely manner?
- Do you have issues with too many users trying to access data at the same time?
- Are you leveraging holistic data from your Google ads spend?

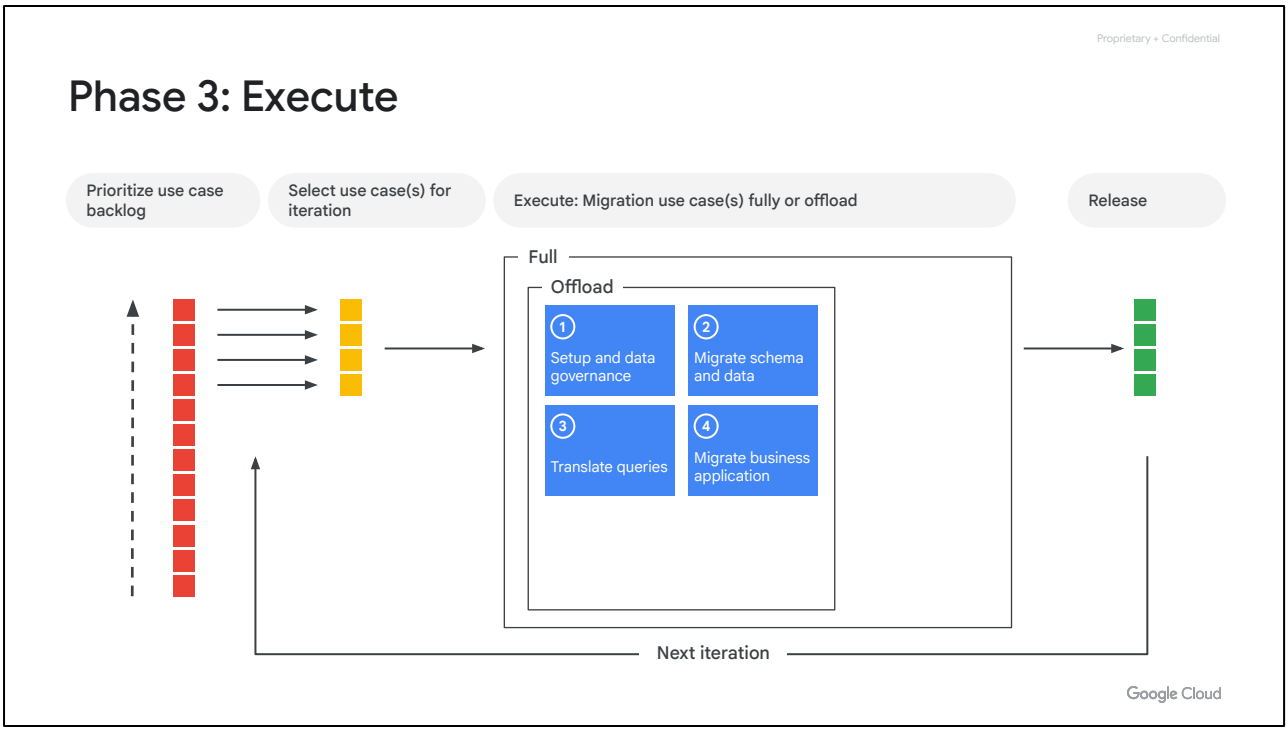
Google Cloud

Many of the things users wish they could do require access to additional or better quality data in a more timely fashion. The migration already involves making changes, so this is an opportunity to identify these requirements and propose them as part of the migration plan. The additional cost will probably be minimal and end users will be happier and much more supportive of the new system. People don’t like change unless it provides them with something they don’t already have!



A plan to migrate using the **offload** method may look something like this. The sources are still feeding the legacy data warehouse, such as Teradata, which is synchronized with BigQuery. Users generate their reports, dashboards and analyses from BigQuery rather than the legacy data warehouse. If new data sources are added during the migration, they feed into BigQuery directly.

This diagram may represent the final result, but typically the migration will progress to a **full** migration, in which the processes that get data from the current sources are modified or replaced to feed BigQuery directly. More on that in a moment.



Now that you've identified and documented the requirements and devised a plan, it's time for action which brings us to the Execute phase. Regardless of the type of migration (offload or full), you need to start the migration by creating a cloud environment. Setup can include the configuration of your Google Cloud projects, network, virtual private cloud (VPC), and data governance.

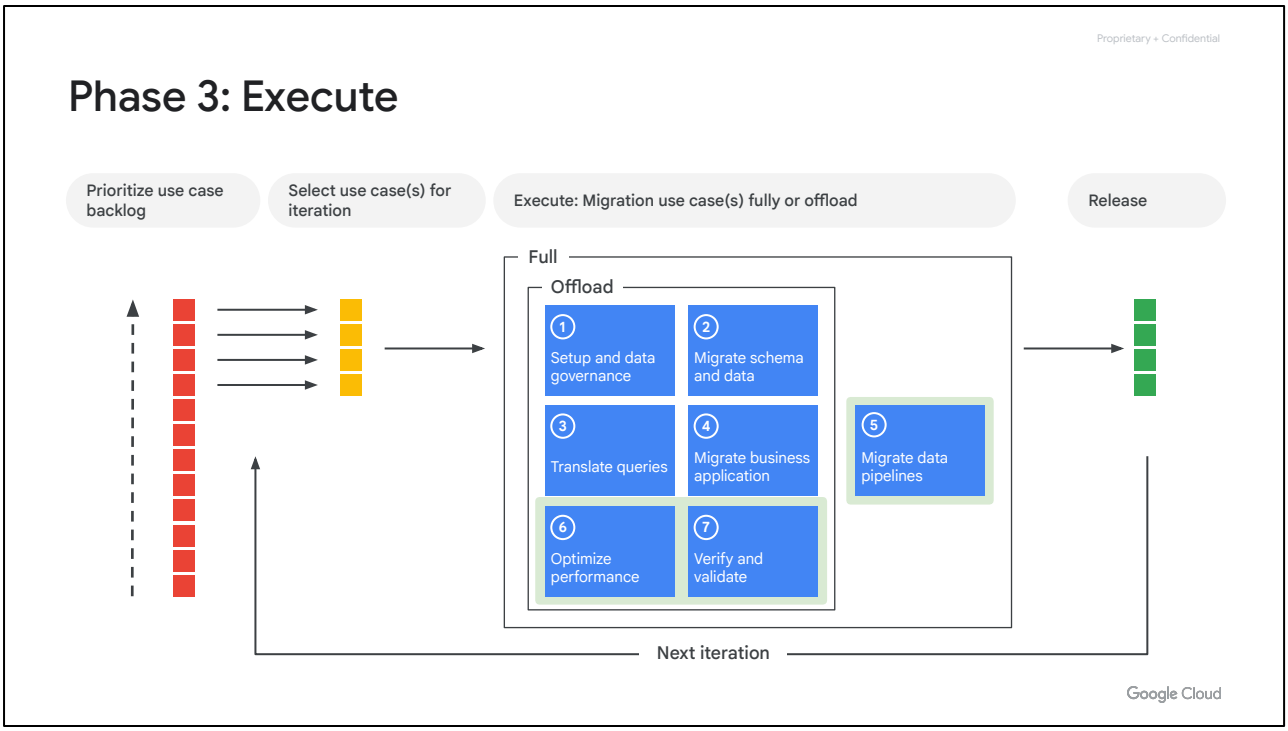
1. Data governance is a principled approach to manage data during its lifecycle, from acquisition, to use, to disposal. A data governance program clearly outlines policies, procedures, responsibilities, and controls surrounding data activities. The program helps to ensure that information is collected, maintained, used, and disseminated in a way that both meets the organization's data integrity and its security needs. It also helps empower employees to discover and use the data to its fullest potential. The data governance document helps you understand data governance and the controls that you need when migrating an on-premise data warehouse to BigQuery.

2. Data is the core of the data warehouse system, so that's the first thing to migrate. The difficulty of this process and the tools that you use depends on whether you're changing the schema. It's simpler to maintain the same schema, but the migration represents a good opportunity to clean up both the data and the schema to eliminate unnecessary information and to optimize the structure of the remaining data. Your customer will have to decide if the increase in time and cost to change the schema is worthwhile.

3. Although SQL is an ANSI standard query language, vendors have added extensions to the language to support their environment. For example,

1. Teradata includes a wide array of extensions, including the ability to create stored procedures to facilitate using Teradata to perform ETL. BigQuery also doesn't support stored procedures, so these processes must be rewritten or replaced during the migration. If this phase is just offloading, only the queries that support users need be revised at this stage. Google has produced a guide to help you with SQL migration from other SQL dialects into BigQuery Standard SQL queries: https://cloud.google.com/bigquery/docs/batch-sql-translator.

2. Business applications in this context refer to dashboards, reports, analytics or pipelines that provide feedback loops, such as the results from a machine learning model. These are the users you identified in Phase 1.



5. At this point, the **offload** migration is complete, so long as the setup included synchronization of the legacy data warehouse with BigQuery. Typically, the ultimate goal is a **full** migration that completely replaces the legacy data warehouse, so at some point you have to migrate the processes that feed the data warehouse. This could be as simple as changing the sink on an existing pipeline if the customer is using something like Informatica, or creating a completely new process if the customer wants to take advantage of new cloud-based technology. We'll talk more about the latter in the **ETL Tools and Positioning** module.

6. In phase 1 and 2, you identified performance requirements and now you need to ensure that you meet or exceed them. BigQuery should perform well without modification in the newly migrated data warehouse. If you find that query performance doesn't match expectations, follow the performance considerations and query optimization strategies discussed in the BigQuery performance optimization documentation.

7. If you followed the guidance in Phase 2 to identify metrics for success and created a test plan, this step should be fairly straightforward. However, it may take quite a bit of effort to tweak the environment. No matter how hard you try, there are always additional queries, pipeline modifications, or other changes that were missed or that require modification to work in the new environment. It's important to document the changes required at this stage, so that you can learn from them and incorporate them in the next iteration of this project and also projects you do in the future.

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Phase 1 of the migration framework described the requirement to understand the current security model, so that you can port it to Google Cloud. In this lesson, we'll discuss options for implementing the various types of security for BigQuery on Google Cloud.

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BigQuery uses Identity and Access Management to manage access to resources

IAM manages access to resources:

- Organizations
- Projects
- Datasets

Roles are assigned to a user, group, or service account to grant access.

Google Cloud

BigQuery uses Identity and Access Management (IAM) to manage access to resources. This is the same security mechanism used across the whole of Google Cloud.

The three types of resources available in BigQuery are organizations, projects, and datasets. In the IAM policy hierarchy, datasets are child resources of projects. A table inherits permissions from the dataset that contains it.

To grant access to a resource, assign one or more roles to a user, group, or service account. Organization and project roles affect the ability to run jobs or manage the project, whereas dataset roles affect the ability to access or modify the data inside a project.

IAM provides two types of roles: predefined and basic. When you assign both predefined and basic roles to a user, the permissions granted are the combined permissions for each individual role.

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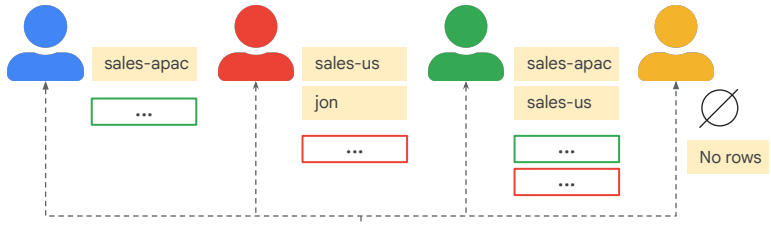
Row-level security in BigQuery

```
CREATE ROW ACCESS POLICY
   apac_filter
ON
   dataset1.table1
GRANT TO
("group:sales-apac@example.com")
FILTER USING
   (Region="APAC");
```

```
CREATE ROW ACCESS POLICY
   us_filter
ON
   dataset1.table1
GRANT TO
("group:sales-us@example.com",
 "user:jon@example.com")
FILTER USING
   (Region="US");
```



Partner	Contact	Country	Region
Example Customers Corp	alice@examplecustomers.com	Japan	APAC
Example Enterprise Group	bob@exampleenterprisegroup.com	Singapore	APAC
Example HighTouch Co.	carrie@examplehightouch.com	USA	US
Example Buyers Inc.	david@examplebuyersinc.com	USA	US

Google Cloud

In BigQuery, row-level security involves the creation of row-level access policies on a target BigQuery table. This policy then acts as a filter to hide or display certain rows of data, depending on whether a user or group is in an allowed list.

An authorized user, with the IAM roles BigQuery Admin or BigQuery DataOwner, can create row-level access policies on a BigQuery table.

When you create a row-level access policy, you specify the table by name, and which users or groups (called the grantee-list) should have access to certain row data. The policy includes the data on which you wish to filter, called the filter_expression. The filter_expression functions like a WHERE clause in a typical query.

In this example, users in the group:apac can only see partners from the APAC region.

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All data stored within BigQuery is encrypted at rest

Google Cloud

An important feature of BigQuery is that all data stored within BigQuery is encrypted at rest. If you want to control encryption yourself, you can use customer-managed encryption keys (CMEK) for BigQuery. Instead of Google managing the key encryption keys that protect your data, you can control and manage key encryption keys in Cloud Key Management Service (Cloud KMS).

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Google Cloud

They say a picture is worth a thousand words, so let's look at potential before-, during-, and after-migration architectures.

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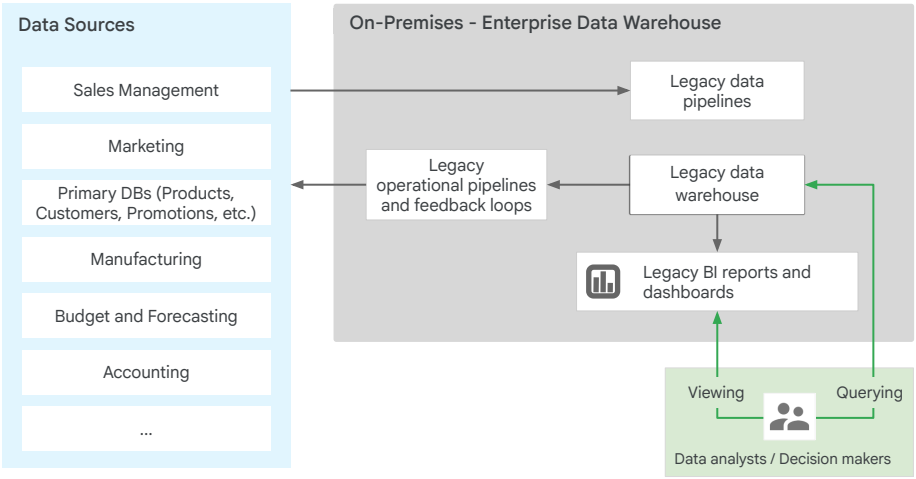
Before the migration: Everything on prem



Google Cloud

Before the migration, all of the data and processes are on-premise. Note that the data sources and the users could be anywhere, but you need to consider how they are delivering or accessing the data, so that you can ensure the data transmissions are secure.

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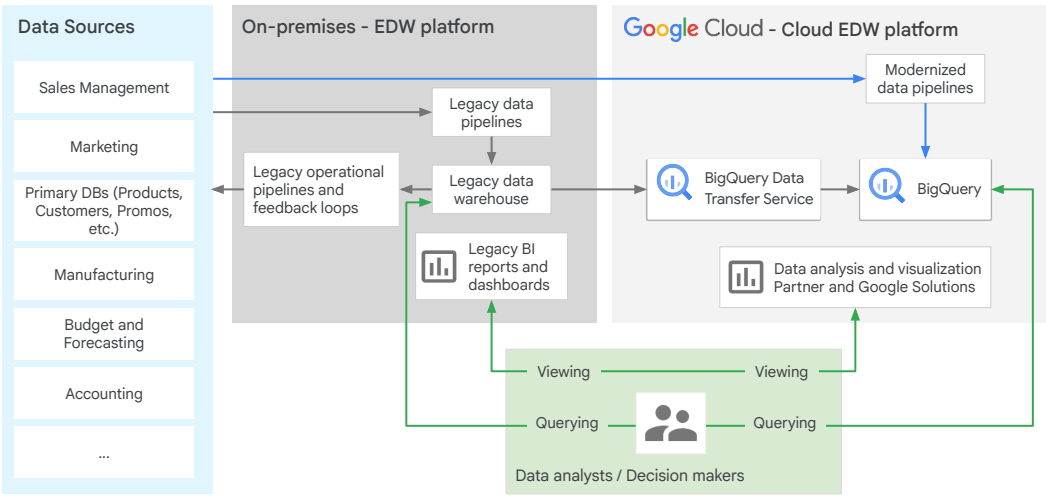
During the migration: Hybrid



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If you recall earlier, I spoke about offloading versus the full migration. Here you have a hybrid architecture which includes both the on-premises data warehouse and BigQuery.

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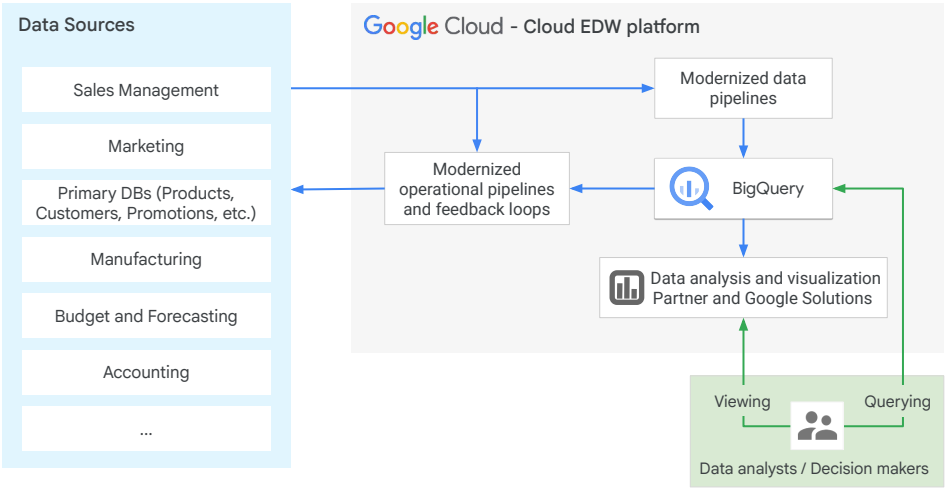
After the migration: Fully cloud



Google Cloud

The Google Cloud implementation looks very similar to the original architecture, except it's now built around BigQuery rather than the legacy data warehouse. It looks similar because BigQuery is a data warehouse and provides the integration necessary for the upstream data pipelines and the downstream reporting, analytical, and BI tools.

BigQuery has additional features including support for geospatial and machine learning functions that you will explore in the **BigQuery Extended Capabilities** module.

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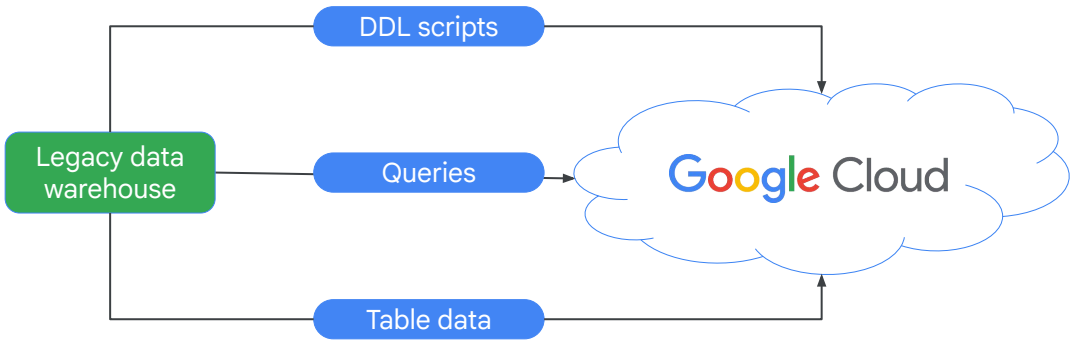
Items to be migrated include tables, views, and scripts



Google Cloud

Let's look a little deeper at some of the assets being migrated.

Migration from a legacy data warehouse, such as Teradata, to Google Cloud involves moving data, scripts, stored procedures, user-defined functions, control flow logic, and the implementation of processes to ensure that data in the new platform can be reconciled with the data on the original one.

So how might you migrate tables and schema?

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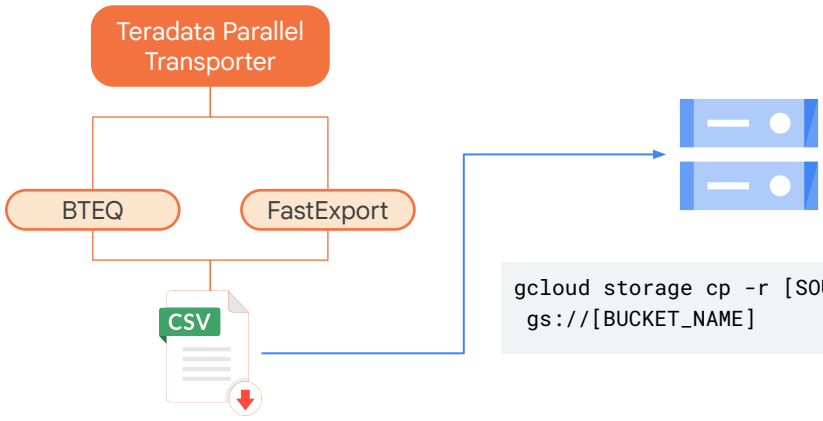
Teradata has grouped its data export utilities into a suite called the Teradata Parallel Transporter




```
gcloud storage cp -r [SOURCE_DIRECTORY] \
 gs://[BUCKET_NAME]
```

Google Cloud

Let's walk through a high-level example using Teradata. One way to migrate data from Teradata is to manually export it using one of the built-in export tools and then copy the data to Cloud Storage. Once the data is uploaded, it can be imported into BigQuery using SQL statements that read directly from Cloud Storage or using a Google Cloud-based ETL tool.

But that requires so much effort; isn't there an easier way?

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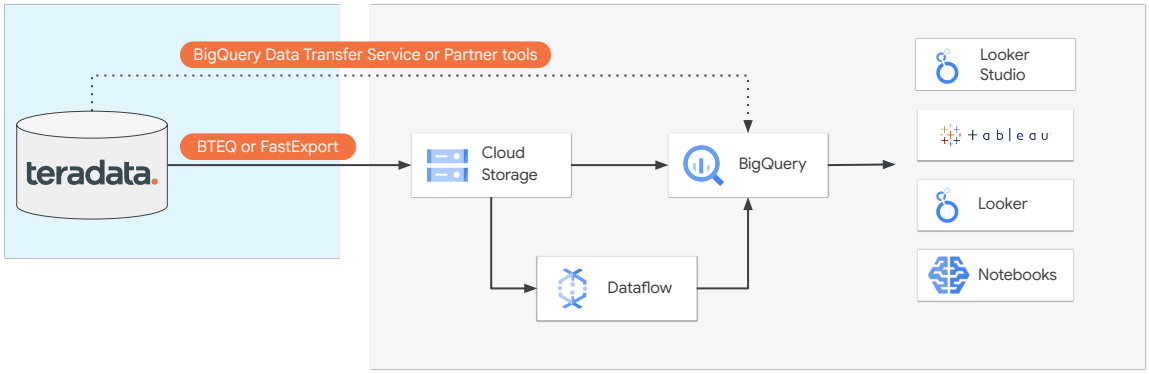
Tools for migrating data from a Teradata Enterprise Data Warehouse to Google Cloud



Google Cloud

Yes, there IS an easier way. The BigQuery Data Transfer Service automates the creation of the current Teradata schema in BigQuery and the initial transfer of data. It can also keep the data synchronized if necessary to simplify the migrate offload approach you looked at earlier. Of course you can also use manual means if you need more control over the process.

Regardless of the approach used, the process is similar. The left-hand side of the diagram shows data being exported from Teradata. Teradata provides two data export utilities capable of sending data from a Teradata platform: BTEQ and FastExport. Alternatively, the BigQuery Data Transfer Service provides a migration service that migrates both schema and data.

Data from Teradata is exported to Cloud Storage, which is the preferred flexible landing place, or staging area, that serves as a repository for the raw data. Cloud Storage is designed for fast, flexible ingestion of structured or unstructured data.

If you don't intend to perform data transformation or the transformation can be accomplished using SQL, you can move data directly from Cloud Storage to BigQuery.

More complex transformations may require services such as Dataflow, which is full managed and serverless. In the **ETL Tools and Provisioning** module, we'll look at Dataflow and other services that could be used to perform transformations.

As indicated by the dotted line, you can do data transformations directly in BigQuery if your team has the requisite SQL skills, and the Teradata administrator will grant you read access to the data. While this approach is technologically feasible, Teradata administrators tend to reject requests to read directly from Teradata.

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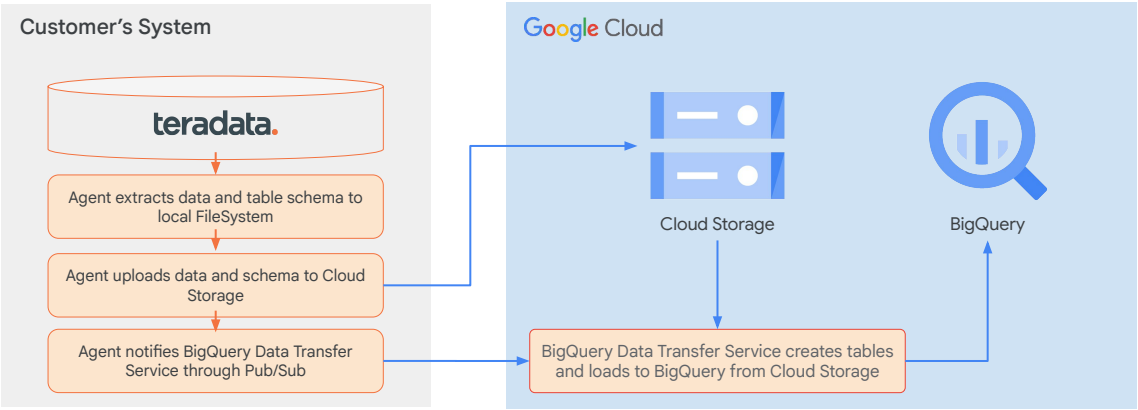
Use the BigQuery Data Transfer Service to copy data from on-premises or cloud storage systems



Google Cloud

The BigQuery Data Transfer Service is the recommended tool for migrating schema and data from Teradata to BigQuery directly. The tool performs initialization and can also perform incremental transfers to keep data synchronized.

The BigQuery Data Transfer Service migrates data only; other database objects, such as stored procedures and views, must be migrated manually.

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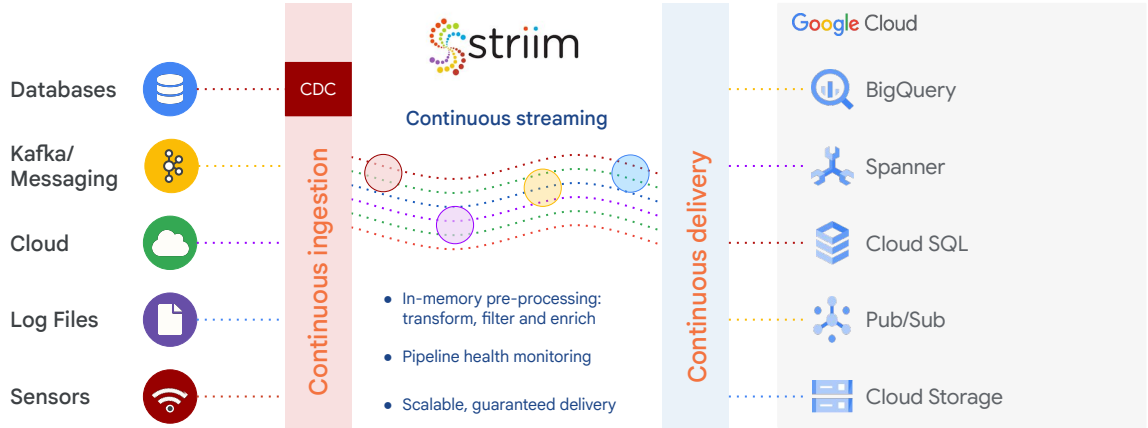
Use Striim to migrate data directly from on-premises or cloud data sources to Google Cloud



Google Cloud

In addition to the migration of a legacy data warehouse, you may find that your customers have other data that they want to migrate to BigQuery.

Striim is a comprehensive streaming extract-transform-load (ETL) platform that enables online data migrations and continuous streaming replication from on-premises and cloud data sources to Google Cloud (including BigQuery, Spanner, and Cloud SQL) through a graphical drag-and-drop interface.

Striims facilitates online database migration to move data from a source database (either on-premises or hosted on a cloud provider) to a target database in Google Cloud, while the source database remains fully accessible to business applications with minimal performance impact.

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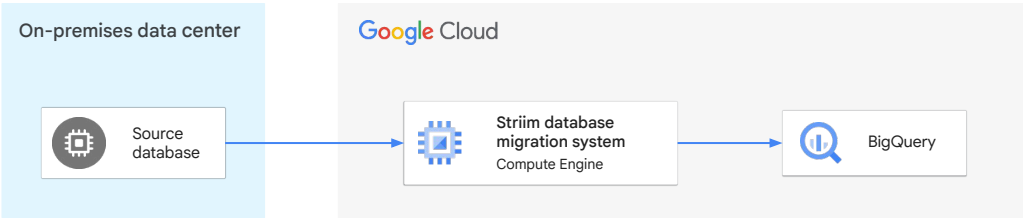
Use Striim to migrate data directly from on-premises or cloud data sources to BigQuery



Google Cloud

For example, you can use Striim to migrate data from an on-premises data source to BigQuery using the following high-level workflow:

1. Create the BigQuery dataset(s) and table(s) to receive the migrated data.
2. Set up a Striim server instance via the Google Cloud Marketplace.
3. Set up a connector between Striim and source database (e.g. JDBC drivers).
4. Configure credentials for Striim to write data to BigQuery (e.g. service accounts).
5. Create and deploy a Striim application for online data migration.

In the hands-on lab for this module, you will apply this workflow to migrate data from SQL Server to BigQuery.

More information on using Striim to migrate data to BigQuery can be found on: https://cloud.google.com/architecture/partners/continuous-data-replication-bigquery-striim

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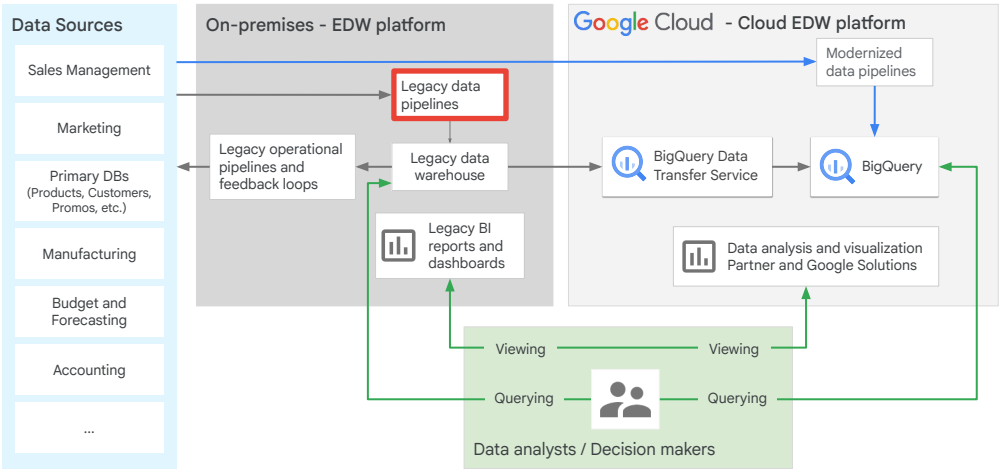
During the migration: Hybrid



Google Cloud

In addition to the data, the data warehouse migration also includes the modification or replacement of data pipelines. In the legacy system, these could be implemented using a variety of methods and tools.

When we looked at the phases of migration, we noted that you could do an offload migration to get up and running using data in the cloud quickly. In this example, the hybrid solution uses the legacy pipelines to update the on-premise data warehouse, and the BigQuery Data Transfer Service keeps BigQuery data synchronized. Over time, the legacy pipelines would be migrated to the cloud.

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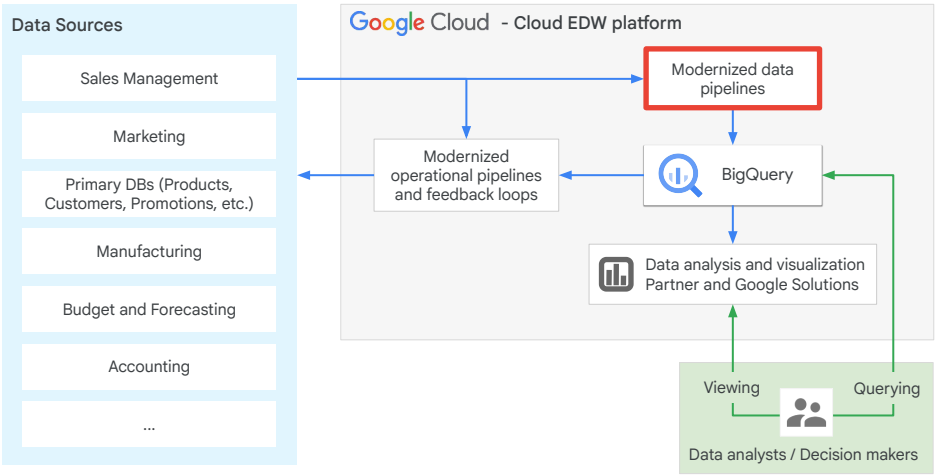
After the migration: Fully cloud



Google Cloud

After all pipelines and processes are migrated to the cloud, the on-premises legacy system can be retired.

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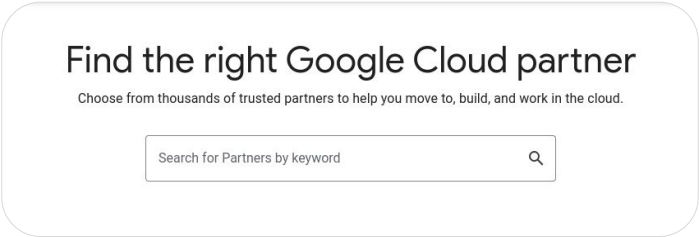
There are several vendors in the ETL space



Google Cloud

So how might you migrate the data pipelines?

In addition to Google Cloud options for implementing new data pipelines such as Dataflow, many ETL market leaders have deeply integrated with both BigQuery and legacy data warehouses such as Teradata and can help extract, transform, load data, and manage processing workflows.

ETL tools have been around for many years. Some organizations might find it convenient to leverage an existing investment in trusted ETL scripts. Some Key partner solutions are included on the BigQuery Solution Providers website.

Knowing when to choose partner tools over Google Cloud native utilities will depend on your current infrastructure and the IT team's comfort level with developing data pipelines in Java or Python code.

https://cloud.google.com/bigquery/providers/

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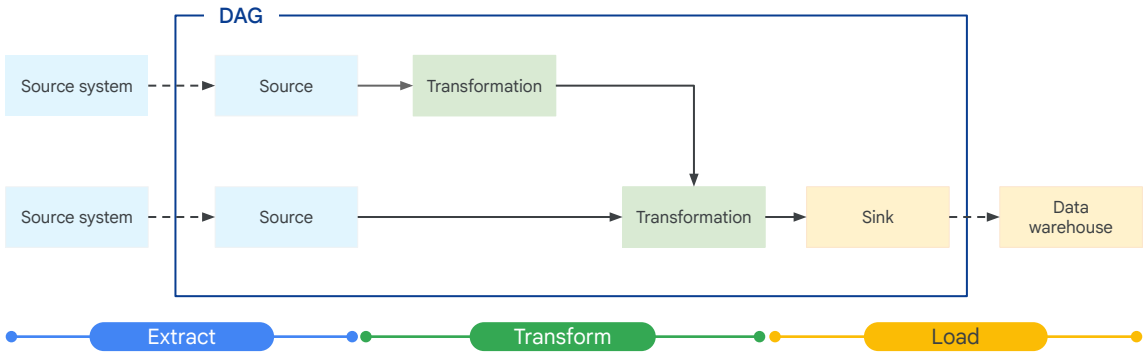
Extract, transform, load (ETL)



Google Cloud

Data pipelines can be implemented in many ways using different tools and methodologies. In a traditional ETL process, the data is Extracted, Transformed, and then Loaded, hence the acronym ETL.

What this means is that the data is extracted in a raw format from the source system and then a process is applied to transform the data into the format in which it's to be loaded into the sink. The transformation might include combining data from multiple sources as part of the process.

Many modern tools take user-written code and produce a DAG (Directed Acyclic Graph) to illustrate the process that will be performed. You will see this in the **ETL Tools and Positioning** module when you explore Dataflow.

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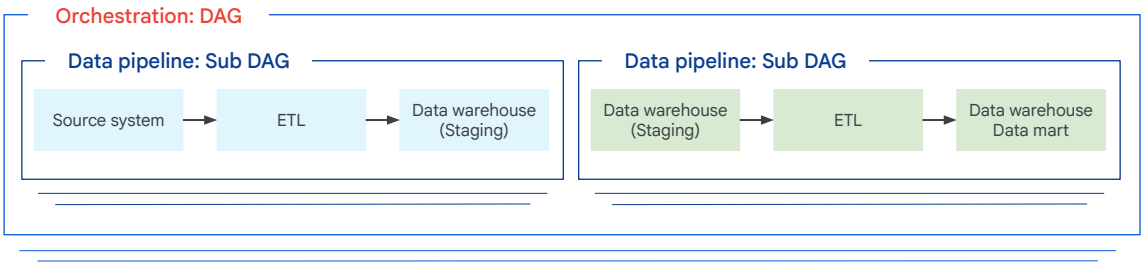
Extract, transform, load (ETL)



Google Cloud

Complex ETL processes may require orchestration to ensure that the processes are completed in a defined order and to enforce dependencies, such as pipeline 1 must complete successfully before pipeline 2 is run.

Cloud Composer is a Google Cloud orchestration tool that can support complex workflows. More on that in the **ETL Tools and Positioning** module.

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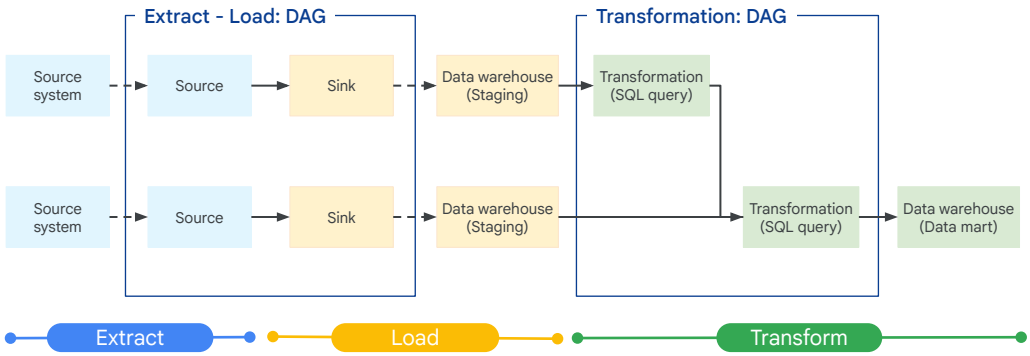
Extract, load, transform (ELT)



Google Cloud

Another type of pipeline is called Extract, Load, and Transform, or ELT. The pipeline accomplishes the same objective of migrating data, however, the transformation is done on the backend.

This approach can take advantage of the parallel processing power in BigQuery, but requires the developers creating the pipeline to write SQL instead of a high-level language.

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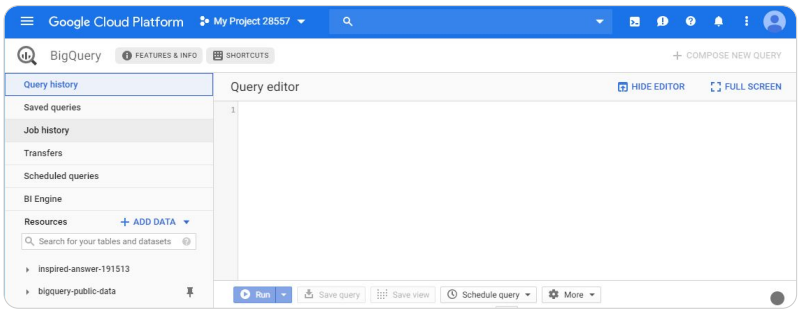
Load and export data, run queries, and perform other user and management tasks in your browser




```
bq query --destination_table=mydataset.topukcustomers \
"SELECT customer_id,customer_name,order_count FROM mydataset.customers WHERE
country = 'UK' ORDER BY order_count DESC LIMIT 6"
```

Google Cloud

The transformation part of the process can be implemented using the BigQuery web UI or programmatically using bq commands at the command line. The bq syntax is easy to learn. Queries can be saved and then executed on a schedule or on demand.

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Keep these data transformation considerations in mind

1. Transform data using BigQuery SQL but understand the BigQuery slot requirements of these jobs.
2. Avoid tying up BigQuery slots for transformation if they interfere with analytical workloads.
3. Schedule expensive, complex transformation jobs with care so they don't interfere with critical analytical queries.
4. Use the Cost Calculator to forecast the cost of BigQuery SQL transformation jobs.
5. Use Dataflow or Cloud Data Fusion to perform ETL jobs.

Google Cloud

There are many cost and performance considerations when choosing a data transformation solution.

1. Using an ELT process takes advantage of the power of BigQuery, but consumes slots…
2. These are the same slots that might otherwise be available for dashboards, reports, and analytical processes.
3. You can reduce the impact on other processes by running the ELT process during off-peak hours or in a lower priority project.
4. If you're not sure of the impact of the transformation process queries, use the estimated cost provided by the BigQuery UI, the bq dry run flag, or the dryfun parameter in an API call.
5. To eliminate BigQuery slot usage altogether, consider using another tool such as Dataflow or Cloud Data Fusion, which are discussed in the **ETL Tools and Positioning** module.

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Product-specific SQL to BigQuery SQL Translation Guide

- High-level migration steps
- Conversion guide
- Conversion reference

https://cloud.google.com/bigquery/docs/batch-sql-translator

Google Cloud

Now that you have ways to transfer your data and data pipelines, how about migrating the SQL scripts?

Google provides documentation to help translate resources and queries in product-specific SQL (such as Teradata) to BigQuery. The reference includes mapping data type, differences in query syntax, and guidance for translating during migration.

There are also third-party products that can support this translation work such as Datametica and Compiler Works.

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Agenda

01 Migration Phases

02 Security

03 Google Cloud Data Warehouse Architecture
Lab: Google Cloud Fundamentals: Getting Started with BigQuery

04 Post Migration

05 User Adoption

06 Summary

Google Cloud

Now, you have the opportunity to complete a hands-on exercise that explores BigQuery.

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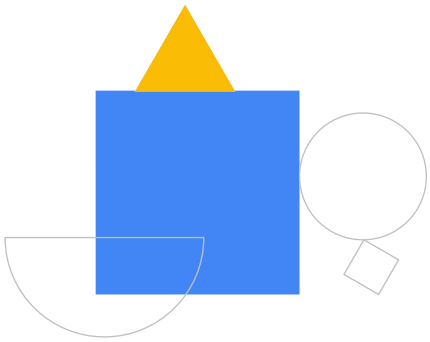
Lab Intro

Google Cloud Fundamentals:
Getting Started with BigQuery



Google Cloud

This hands-on lab demonstrates how to load a web server log into a BigQuery table. After loading the data, you query it using the BigQuery web user interface and the BigQuery CLI.

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Lab objectives

01 Load data from Cloud Storage into BigQuery.

02 Perform a query on the data in BigQuery.

Google Cloud

In this lab, you will learn how to:

- Load data from Cloud Storage into BigQuery.
- Perform a query on the data in BigQuery.

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Google Cloud

After the data migration is complete, you can begin to optimize your usage of the platform to start solving business needs. This might include utilizing the explore and visualize tools of Google Cloud to derive insights for business stakeholders, optimizing under-performing queries, or even developing a program to aid user adoption.

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Data visualization tools can help to make sense of BigQuery data and analyze the data interactively

Looker

Looker Studio

Notebooks

Google Cloud

Users can use visualization tools such as Looker, Looker Studio, and Notebooks to help them identify trends, respond to them, and make predictions using their data.

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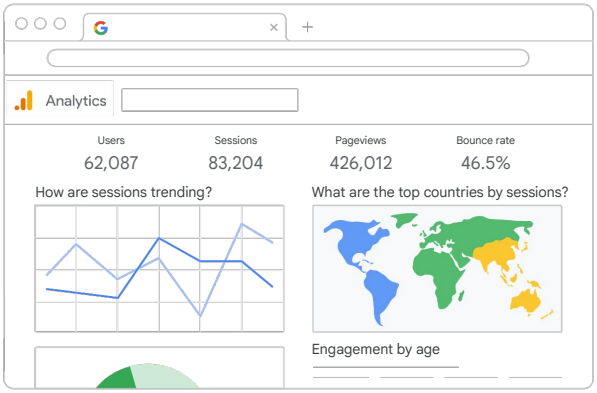
Looker Studio is integrated in Google products and applications



Google Analytics

Billing dashboard

Google Cloud

Looker Studio dashboards are widely used across many Google products and applications.

- For example, Looker Studio is integrated into **Google Analytics** to help visualize, in this case, a summary of a marketing website. This dashboard visualizes the total number of visitors through a map, compares month-over-month trends, and even displays visitor distribution by age.
- Another Looker Studio integration is the **Google Cloud billing dashboard**. You might be familiar with this from your account. Maybe you’ve already used it to monitor spending, for example.

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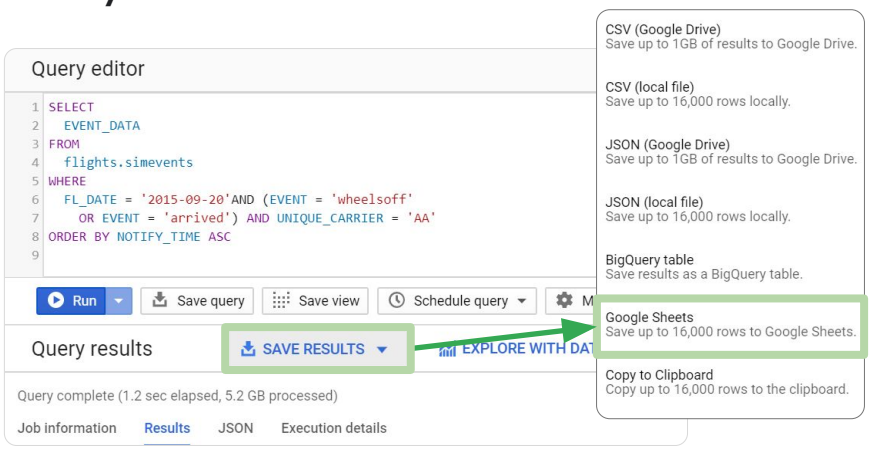
Save query results directly to Google Sheets from the BigQuery UI



Google Cloud

BigQuery integrates with Google Drive, which allows you to save query results directly to Google Sheets from the BigQuery UI. Users will see a “Save to Google Sheets” button in the BigQuery user interface. Clicking on this button saves query results to Google Sheets and prompts you to open that spreadsheet. You can also query Google Sheets spreadsheets from BigQuery as you edit them in Sheets.

Google Cloud partners and third-party developers have developed multiple integrations with BigQuery, making it easy to load, process, and make interactive data visualizations.

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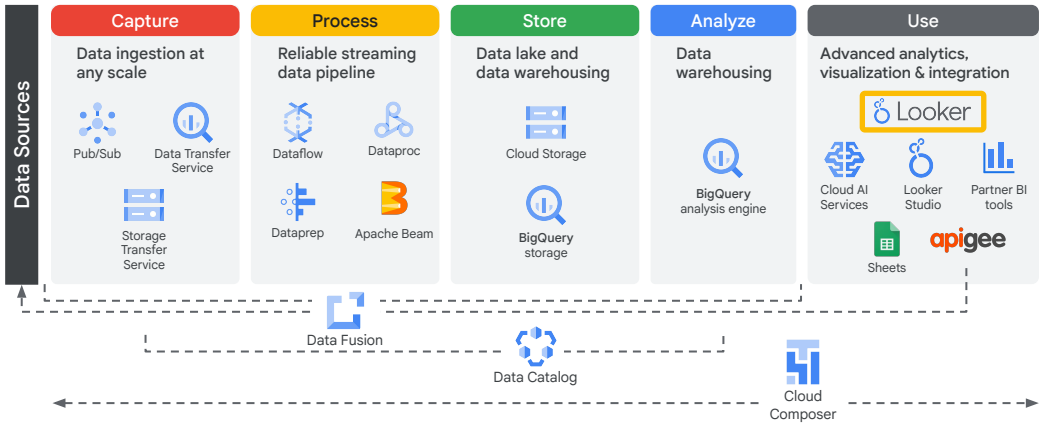
Looker is the Google Cloud enterprise platform for data experiences



Google Cloud

Some customers can benefit greatly from leveraging Looker, Google Cloud's enterprise platform for data experiences — from internal business intelligence to externally embedded analytics to custom data applications.

Looker's unique architecture provides a rich development framework that is built to support enterprise-grade workflows and use cases.

With Looker, organizations can drive better outcomes, enable smarter data experiences, and find true power in their data. Organizations can move beyond traditional, stale BI and infuse fresh data into every workflow.

For more information about BigQuery, refer to the product page: https://cloud.google.com/bigquery.

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Notebooks makes it easy to go from data ingestion to preprocessing and exploration, and eventually model training and deployment

Google Cloud

Notebooks (Vertex AI and AI Platform) is a managed service that offers an integrated JupyterLab environment. ML developers and data scientists can create instances running JupyterLab that come pre-installed with the latest data science and ML frameworks in a single click. Notebooks is integrated with BigQuery, Dataproc, and Dataflow, making it easy to go from data ingestion to preprocessing and exploration, and eventually model training and deployment.

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Google Cloud

To make the transition to BigQuery successful, you need to focus on the people side of this migration, as well as the technical side. Even if you aren't an expert in change management, becoming familiar with some of those tools—the concepts, steps, and best practices—can help you ensure a smooth and complete transition to BigQuery. And correspondingly, users who are happy and productive in the new environment.

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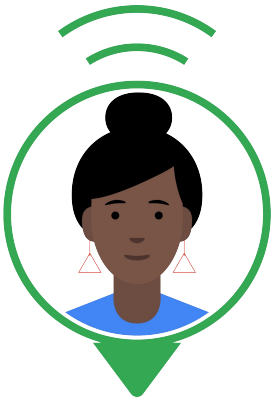
Change management practices provide a structured approach

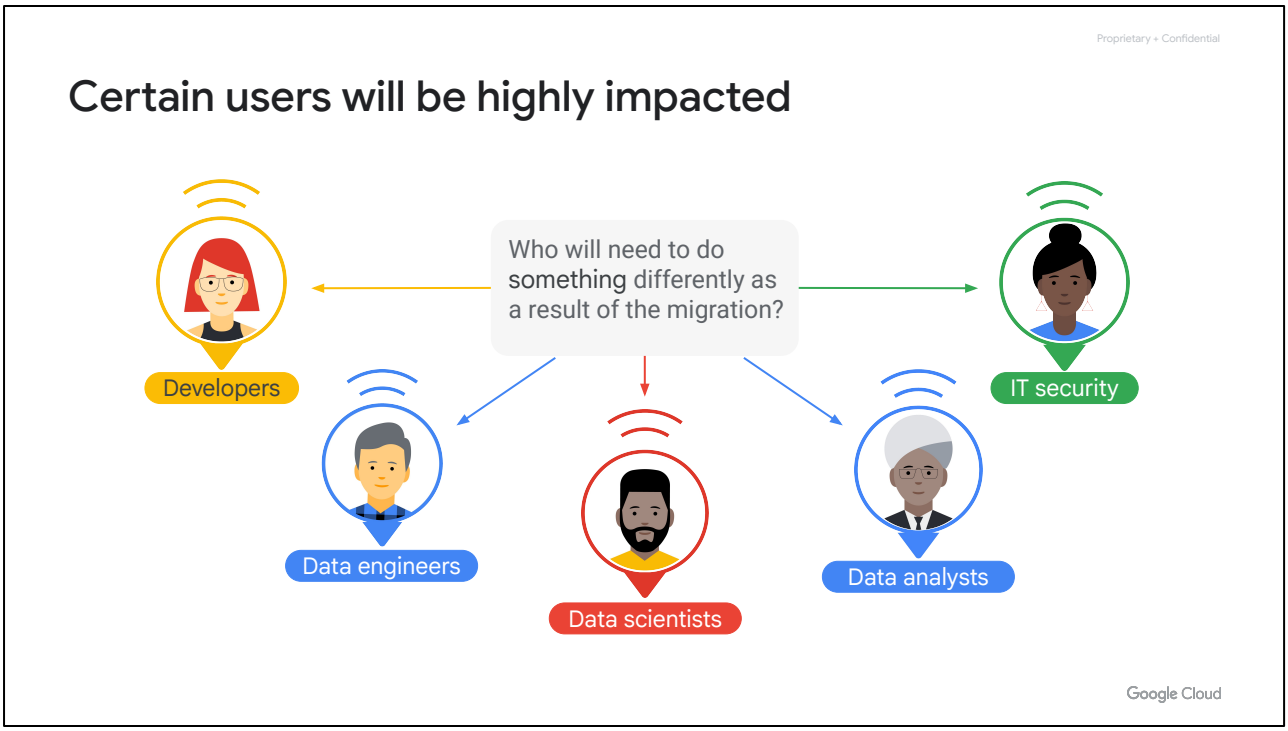


- Why are we switching?
- What's happening to my dashboards?
- How will I get up to speed?

Google Cloud

Change management practices provide a structured approach that will enable change agents to expedite adoption, increase productivity, and address user concerns.

Readiness activities include engaging a sponsor and analyzing the user impact of the migration. Active and visible sponsorship is the most important factor for success in any change management initiative. The sponsor should be an executive, such as the CIO or CTO, who will act as champion for the migration across the organization.



In most organizations, highly impacted users include data architects, developers, data scientists, data analysts, and IT security. To identify other possible users, the question to ask is: Who will need to do something differently as a result of the migration? If the answer isn't known, one should consider sending a survey to employees to discover who will be impacted and how. Change agents should use this information to identify changes required and to develop a plan to support these users throughout the migration.

Some user groups will be highly impacted by the migration to BigQuery.

- **Data engineers** design, build, maintain, and troubleshoot data processing systems with a particular emphasis on the security, reliability, fault-tolerance, scalability, fidelity, and efficiency. Data engineers work in all aspects of the transition to BigQuery, including migrating, updating, and transforming data.
- **Developers** build and create the organization's software applications. They are impacted by the migration because they need to understand changes to data sources and the data model. Developers will also need to know about the new APIs available with BigQuery.
- In many organizations, **data scientists** are the individuals who acquire, transform, explore, analyze, and visualize data. Some data scientists might even build machine learning models and run experiments. With BigQuery, data scientists may need to understand a new, denormalized data model and find new ways to optimize their queries. In addition to BigQuery, they'll also need to learn some new tools (like Vertex AI and Looker Studio) for exploring and visualizing data.

- **Data analysts** run basic queries and reports as well as summarize and share data with others. In BigQuery, business users need to understand how to get the data they want using the BigQuery web UI. More importantly, business users need to have an easy way to visualize the data so they can share it through presentations, dashboards, or reports.
- The **IT security** team has a strong understanding of industry best practices around information security. The team helps perform security assessments, code audits, and design reviews. IT security needs to ensure proper authentication, authorization, and data access in BigQuery, as well as the overall security of the Google Cloud environment.

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A proactive approach to communication with users is critical to success

Migration messages

- Migration site
- Migration announcements

Events

- Big data hackathons
- Big data demo days

Office hours

- Open sessions

Google Cloud

A proactive approach to communication with users is critical to success. There are some suggested ways to communicate the migration to users.

- A migration site is a centralized site containing information about the migration including the scope, schedule, Frequently Asked Questions, and announcements. Migration announcements can be used to announce the current status of the migration, remind users of timing, and inform users about migration tasks.
- Big Data hackathons are collaborative events where users are encouraged to build new queries, analyses, and visualizations using BigQuery. They can be used to make the process of migrating more fun. Big Data demo days are live events where different users or teams showcase examples of how they're using BigQuery. These can motivate users to produce queries that they can “show off” to their colleagues.
- Change agents may decide to schedule other types of events on a regular basis to keep users engaged and focused on the new system. Open sessions are scheduled times when users can come and ask questions related to BigQuery.

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Training can take many forms for BigQuery tools

Introduction to AI and Machine Learning on Google Cloud	BigQuery for Data Analysts	Data Engineering on Google Cloud	Skill badges for Data Engineers	Google Certified Data Engineer
8-hour ILT Introduces Google Cloud's AI and ML capabilities, with a focus on developing both generative and predictive AI projects.	3-day ILT Hands-on skills in ingesting, analyzing, transforming, and visualizing data using BigQuery and getting the best performance when using it at scale	4-day ILT Hands-on introduction to designing and building data processing systems on Google Cloud	Online collection of hands-on labs and learning materials related to mastery of data engineering	Certification recognizes IT professionals with the exceptional ability to design, build, and maintain big data and machine learning solutions on Google Cloud
Data Engineers Data Scientists Data Analysts	Data Scientists Data Analysts	Data Engineers	Data Engineers	Data Engineers

Google Cloud

BigQuery is easy to use, but everyone learns at their own pace and learns new things differently. So how exactly can you make sure that the needs of users are met? This outline is an approach that an organization can use to ensure users can be productive quickly.

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01	Migration Phases
02	Security
03	Google Cloud Data Warehouse Architecture **Lab:** Google Cloud Fundamentals: Getting Started with BigQuery
04	Post Migration
05	User Adoption
06	Summary

Google Cloud

That concludes the ‘Migrating to BigQuery’ module. As you have seen, success in migrating a data warehouse system involves a lot of concerted effort. Let’s briefly review the process.

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Summary

- The migration framework helps you structure, plan and execute your migration.
- During the planning phase, perform the due diligence required to successfully deliver the business use case.
- Offloading data and schema from your legacy data warehouse to BigQuery is a great way to migrate and parallelize efforts.
- The last phase of migration focuses on ensuring source systems and data pipelines are enhanced to feed directly into Google Cloud.

Google Cloud

- Every major undertaking should have a process associated with it. We looked at a migration framework that can serve as a roadmap for migrating a data warehouse. Having a roadmap and checklists helps prevent missing elements in the migration.
- It's necessary to turn over every rock to ensure that you identify all of the requirements and accurately determine the scope of the effort required for each use case.
- Data is the core of a data warehouse, so you can get something up and running quickly using an offload migration rather than full migration. Offloading means that you move the schema and data to the cloud and then point end users to the migrated data.
- If you started with an offload migration, the final step will be to modify or replace the existing pipelines so that they point to BigQuery instead of the legacy data warehouse.

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Summary

- BigQuery uses IAM to manage access to resources and provides additional levels of security (table, row, column).
- Change management practices provide a structured approach to the migration process.

Google Cloud

- BigQuery uses the same security model as all other services in Google Cloud, so no separate administrator is required to manage security at the project or dataset level. You can also implement table, row-level, and column-level security in BigQuery.
- Many people are averse to change, so it’s incumbent on you to understand and manage the impact of the new system to ensure a smooth transition.